

CMPSC 480 Midterm Examination

Cumulative assessment of Weeks 1-7

CMPSC 480 - Student Question Deck

Assessment overview

Solve applied, mathematical, algorithmic, and engineering problems under explicit assumptions.

Coverage: Modules A and B: agents, search, adversarial reasoning, MDPs, RL, approximation, and Bayesian inference

Points

100 total

Expected time

110 minutes

Submission

Individual examination PDF

Learning objectives

- ▶ Formulate agent and search problems precisely.
- ▶ Trace cost-sensitive and adversarial algorithms.
- ▶ Apply Bellman and temporal-difference updates.
- ▶ Reason about function approximation stability.
- ▶ Compute Bayesian posteriors and factor operations.
- ▶ Integrate methods while preserving assumptions.

Requirements checklist

- ▶ Work independently and show intermediate algebra, frontier state, or factor state.
- ▶ One double-sided handwritten letter-size reference sheet is permitted.
- ▶ A nonprogrammable calculator is permitted; networked devices and communication are prohibited.
- ▶ State every tie-breaking, terminal, reward, and zero-division convention you use.
- ▶ Answers receive credit for justified reasoning; an unsupported final number may receive little credit.
- ▶ Academic integrity rules of the course and institution apply throughout the examination.

Task 1 - Agent and search formulation (10 points)

A delivery robot operates in a partially mapped building with battery limits and moving people.

- a. Give one concrete element for each PEAS component. [3 points]
- b. Define states, actions, transition model, goal test, and path cost for deterministic route planning. [3 points]
- c. Explain why rational behavior does not imply omniscience and name two defensive interface checks. [4 points]

Task 2 - UCS and A-star (15 points)

Graph edges are S-A:4, S-B:1, B-A:1, A-G:3, and B-G:8. Heuristic values are $h(S)=5$, $h(A)=3$, $h(B)=4$, $h(G)=0$.

- a. Trace UCS until a goal is removed, including cheaper rediscovery. [5 points]
- b. Trace A-star priorities and return the selected path. [5 points]
- c. Is the heuristic admissible and consistent? Justify from the graph. [5 points]

Task 3 - Minimax and alpha-beta (15 points)

A MAX root has MIN children A, B, and C. Their leaves in search order are A:[6,4], B:[3,8], C:[5,2].

- a. Compute every minimax value and the selected action. [5 points]
- b. Trace alpha-beta left to right and identify pruned leaves. [5 points]
- c. Design a depth-cutoff evaluation with three features and state two tests. [5 points]

Task 4 - MDP planning (15 points)

At state s , action a reaches terminal reward 5 with probability 0.8 and state t with reward 0 and probability 0.2. Let $\gamma=0.9$ and $V(t)=2$. Action b deterministically yields reward 3 and terminates.

- a. Compute $Q(s,a)$ under reward-on-transition semantics. [5 points]
- b. Compute $Q(s,b)$ and the greedy action. [5 points]
- c. Explain why value iteration converges for a finite discounted MDP. [5 points]

Task 5 - Q-learning and approximation (15 points)

A linear agent has $w=[0.5,-0.5]$, $f(s,a)=[1,2]$, reward 2, $\gamma=0.5$, max next $Q=3$, and $\alpha=0.2$.

- a. Compute current Q , target, TD error, and new weights. [5 points]
- b. Recompute the update if the transition is terminal. [5 points]
- c. Name the deadly triad and two observable instability signals. [5 points]

Task 6 - Bayesian networks and inference (15 points)

R is a binary root with $P(R)=0.2$. S depends on R with $P(S|R)=0.1$ and $P(S|\text{not } R)=0.5$. W depends on both with $P(W|R,S)=0.99$ and $P(W|R,\text{not } S)=0.8$.

- a. Compute $P(R,S,W)$ for all three true. [5 points]
- b. Write an exact expression for $P(R|W)$ without evaluating missing CPT rows. [5 points]
- c. If factors $f(R,S)$ and $g(S,W)$ are multiplied and S is summed out, what is the result scope and why can order matter? [5 points]

Task 7 - Integrated engineering scenario (15 points)

A robot uses a Bayesian belief over hazards as features for a linear Q-learning policy.

- a. Specify the belief-to-policy interface and three invariants. [5 points]
- b. Design a controlled ablation showing whether belief tracking helps. [5 points]
- c. State a response to impossible evidence and a posttraining safety gate. [5 points]

Edge cases and defensive programming

- ▶ A start state is already a goal.
- ▶ All minimax root actions tie.
- ▶ A terminal RL transition has a large next Q.
- ▶ Bayesian evidence has zero probability.
- ▶ A factor order creates an unexpectedly large table.

Submission instructions

- ▶ Write your name and identify every requested problem and subpart.
- ▶ Show enough intermediate work that another reader can audit the result.
- ▶ Submit the complete question-response document when time is called.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric - part 1

Category	Points	Required evidence
Agent and search formulation	10	PEAS analysis, search model, and engineering contract.
UCS and A-star	15	Priority trace, optimal path, and heuristic argument.
Minimax and alpha-beta	15	Minimax backups, pruning trace, and evaluation design.
MDP planning	15	Bellman values, policy choice, and convergence reasoning.

Grading rubric - part 2

Category	Points	Required evidence
Q-learning and approximation	15	Semi-gradient update, terminal change, and stability analysis.
Bayesian networks and inference	15	Joint probability, posterior structure, and factor operation.
Integrated engineering scenario	15	Interface, ablation, and safety verification plan.

Reflection prompt

In one or two sentences:

Identify one assumption that changed the answer to a problem.